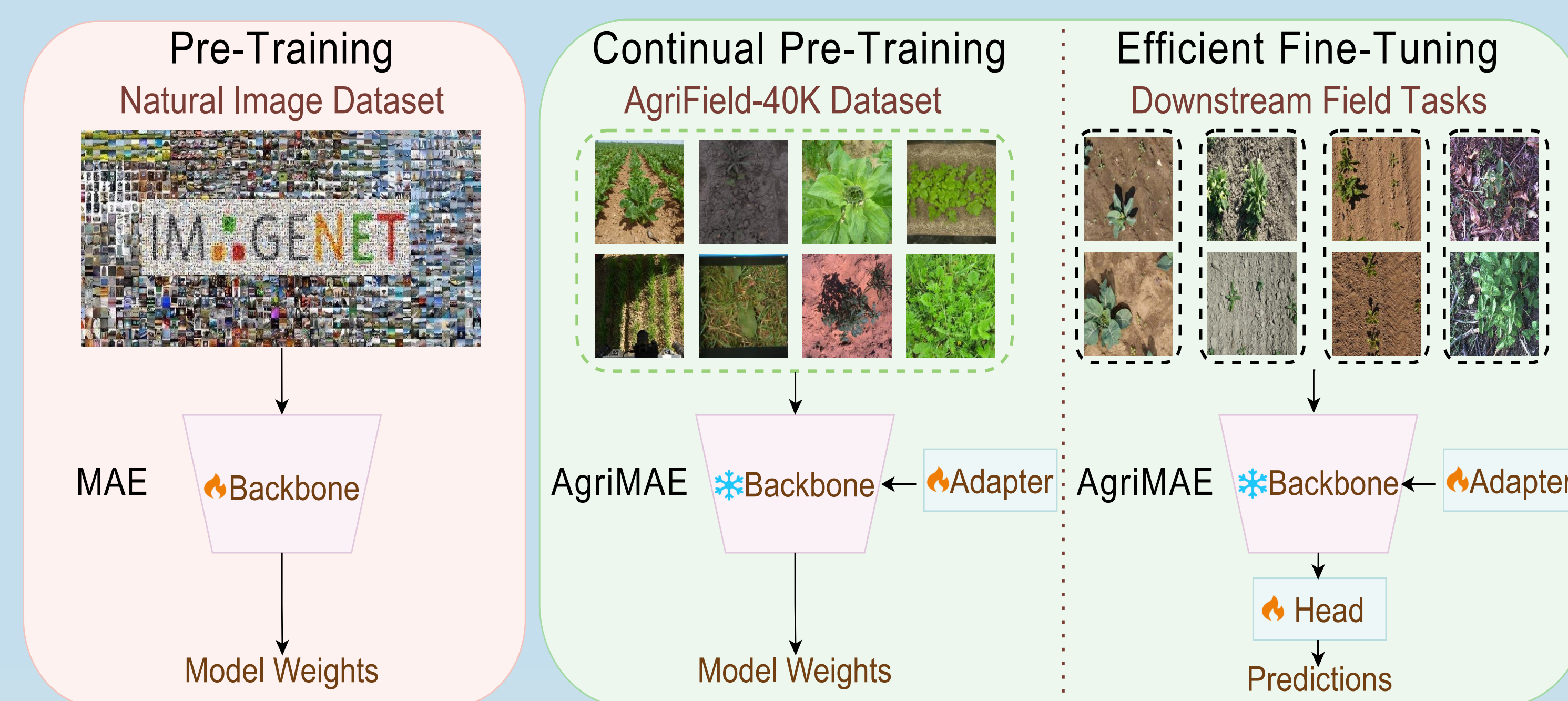


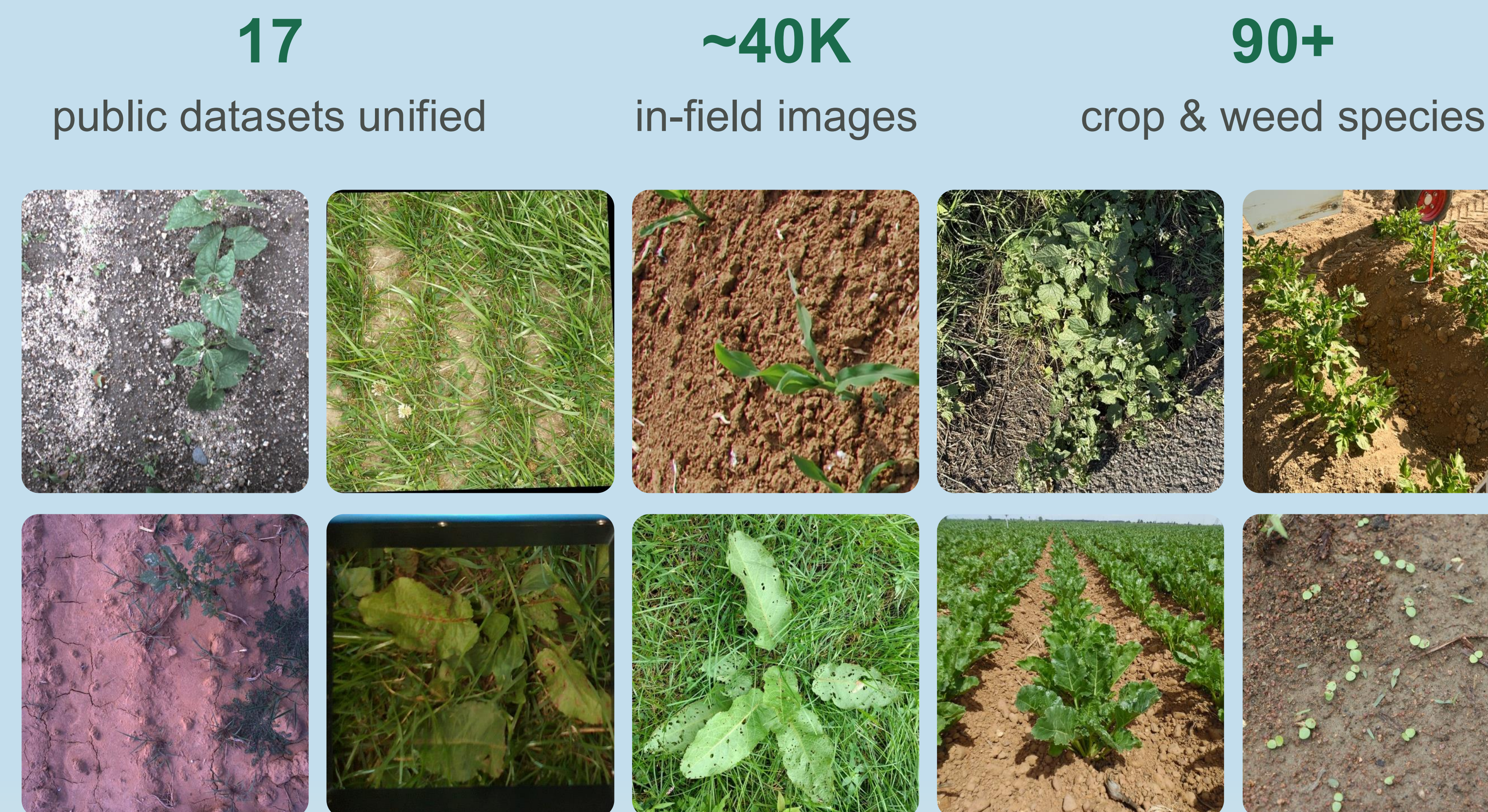
Introduction

- ✗ Natural-image pretrained models miss the agricultural domain
- ✗ Full continual pretraining is too expensive
- ✗ Agricultural datasets are small and scattered
- ✓ **Parameter-efficient continual pretraining**
- ✓ **One unified in-field corpus of 40K images**

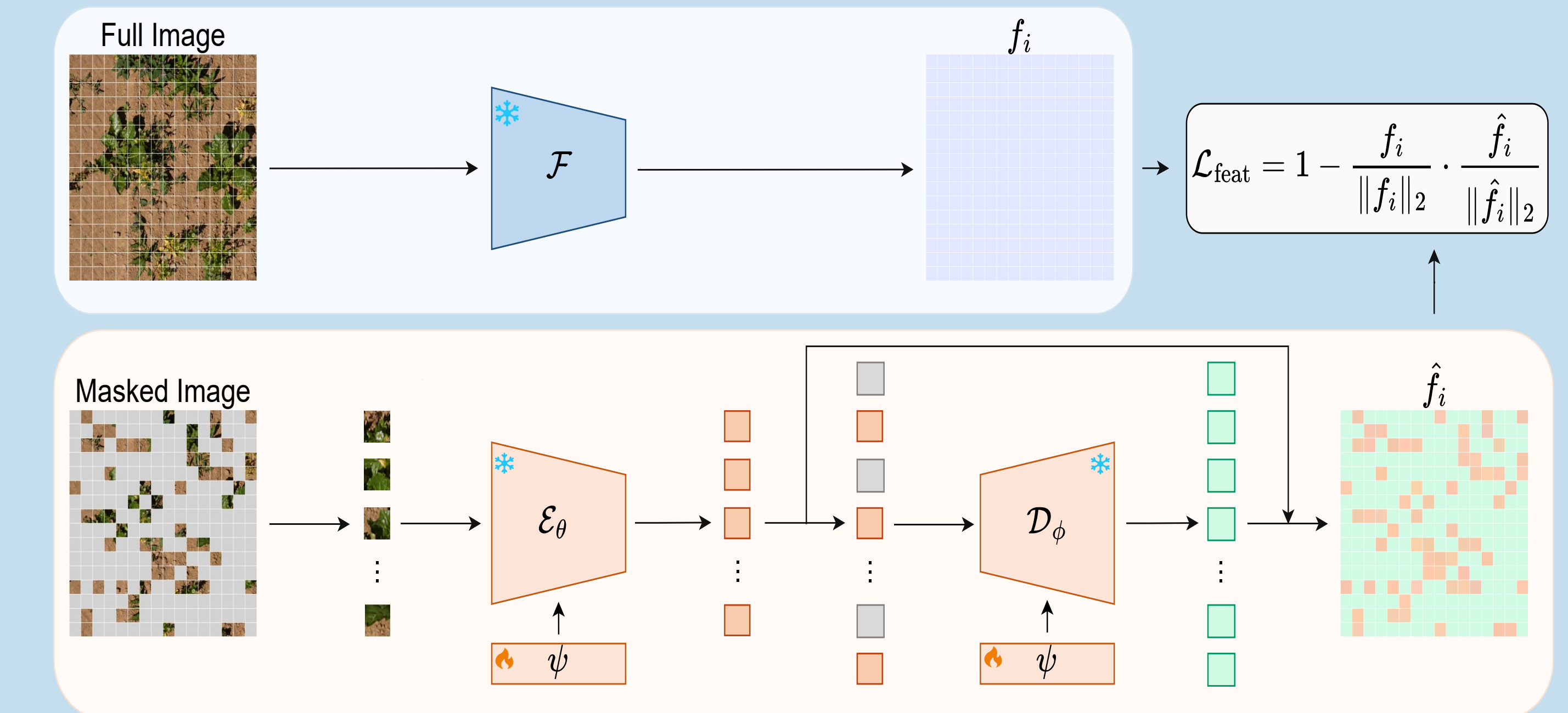


AgriField-40K

A field-centric corpus for continual pretraining



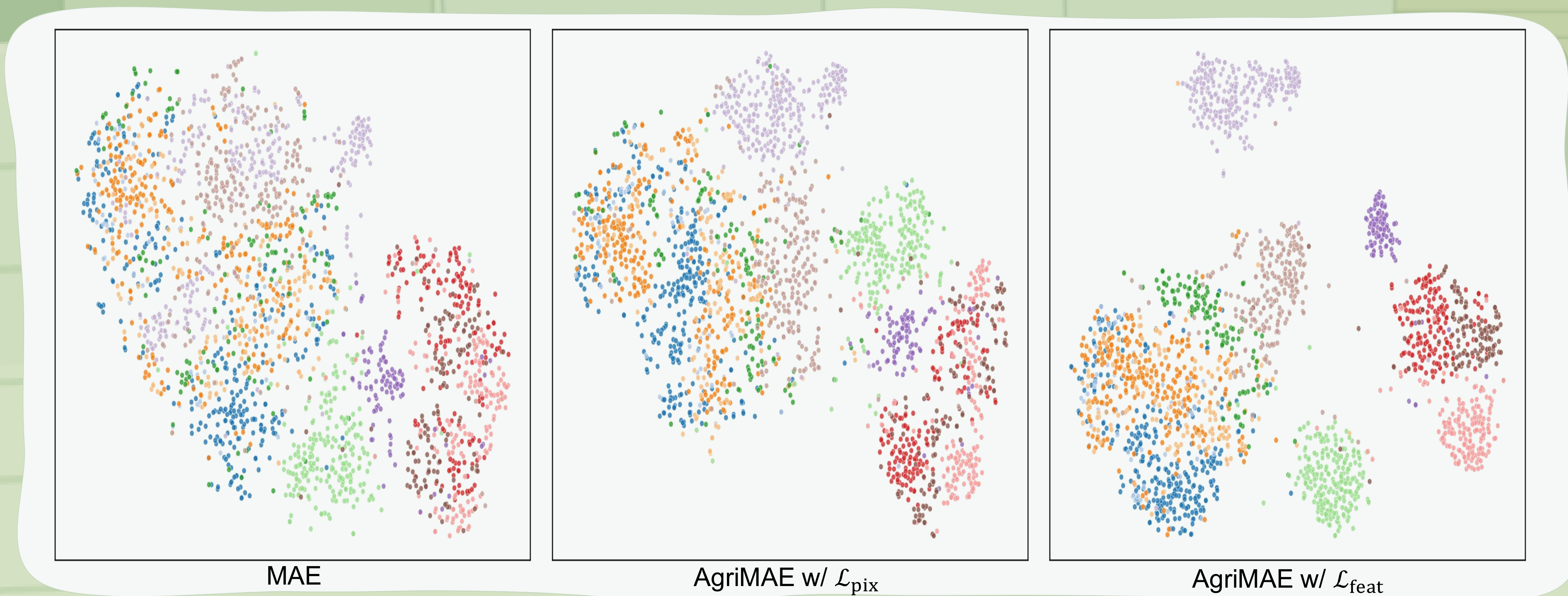
AgriMAE



- Pixel ($\mathcal{L}_{\text{pix}}$) vs. semantic feature ($\mathcal{L}_{\text{feat}}$) reconstruction
- Only lightweight adapters are trained
- Adapters reused across downstream tasks

Representation Quality

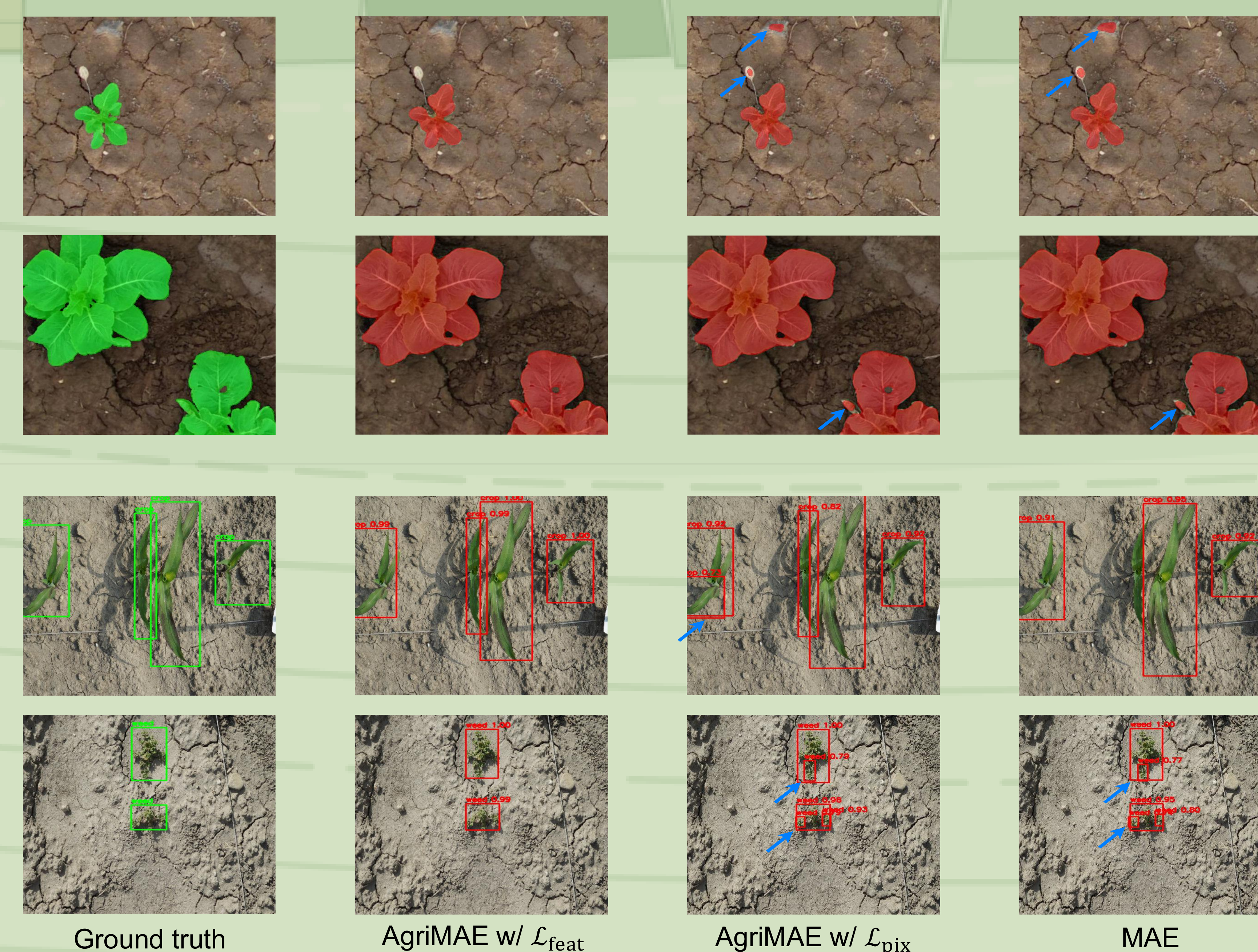
Semantic targets improve separability



- **MAE (ImageNet weights)** — mixed clusters, no crop identity
- **Pixel-level (AgriField-40K)** — better separation, noisy boundaries
- **Feature-level (AgriField-40K)** — clear, semantically consistent clusters

Results

Matches or beats full fine-tuning at a fraction of the training cost



+2.5 pts

DeepWeeds top-1: 92.2% → 94.7% (~9x fewer params)

+3.0 pts

CropAndWeed mAP50: 63.3% → 66.2% (~4x fewer params)

+0.1 pts

PhenoBench weed IoU: 57.1% → 57.3% (~3x fewer params)

97% of full FT

GrowliFlower mIoU: 58.1% vs 59.9% (~3x fewer params)